



ANDERSON SERANGOON JUNIOR COLLEGE
HIGHER 1

2021 JC2 PRELIMINARY EXAMINATION

BIOLOGY

8876/01

Paper 1 Multiple Choice

21 September 2021
Tuesday

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and identification number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **19** printed pages and **1** blank page.

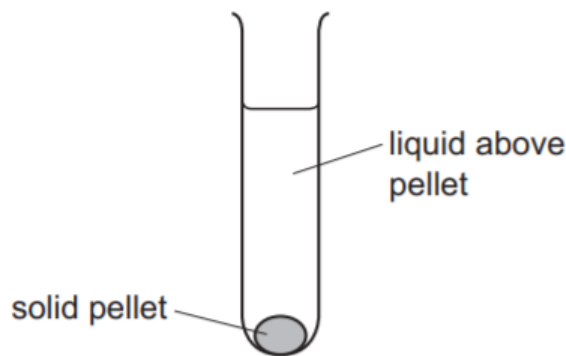
1 Which cellular structures are typically found in a bacterial cell?

- A 30S ribosomal subunit, cell surface membrane, ligase
- B 40S ribosomal subunit, secretory vesicle, protease
- C 50S ribosomal subunit, mitochondrion, DNA polymerase
- D 60S ribosomal subunit, circular DNA, ATP synthase

2 A scientist carried out an experiment to separate the organelles in an animal cell by mass.

The scientist mixed the cells with a buffer solution which had the same water potential as the cells. He then broke the cells open with a blender to release the organelles.

The extracted mixture was filtered and then spun in a centrifuge at a speed to separate the heaviest organelle. This sank to the bottom, forming a solid pellet, 1.



The liquid above pellet 1 was poured into a clean centrifuge tube and spun in the centrifuge at a higher speed to separate the next heaviest organelle. This organelle sank to the bottom, to form a solid pellet, 2.

He repeated this procedure twice more to obtain pellet 3 and pellet 4, each containing a single organelle.

What is the function of the organelle extracted in pellet 3?

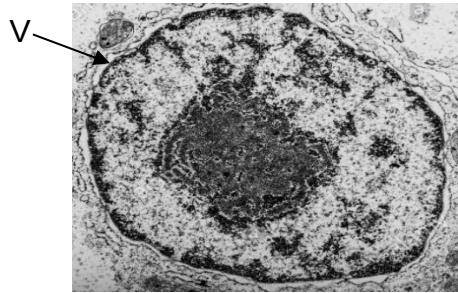
- A production of ATP
- B synthesis of mRNA
- C synthesis of protein
- D digestion of old organelles

- 3 Four students were asked to match the function with the appearance of some cell structures in an animal cell.

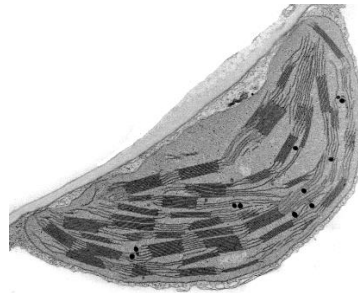
The functions were listed by number.

- 1 mRNA passes through to the ribosome
- 2 detoxifies metabolic wastes
- 3 aids microtubule formation

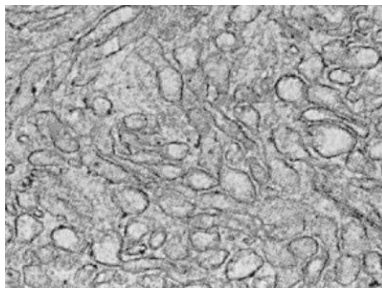
The appearances (not to relative scale) were listed by letter.



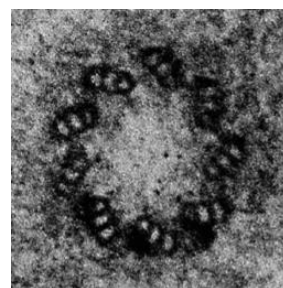
Y



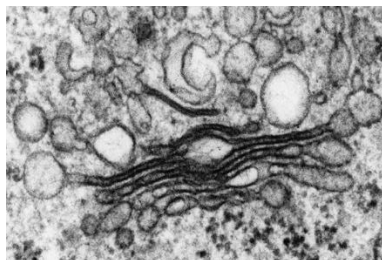
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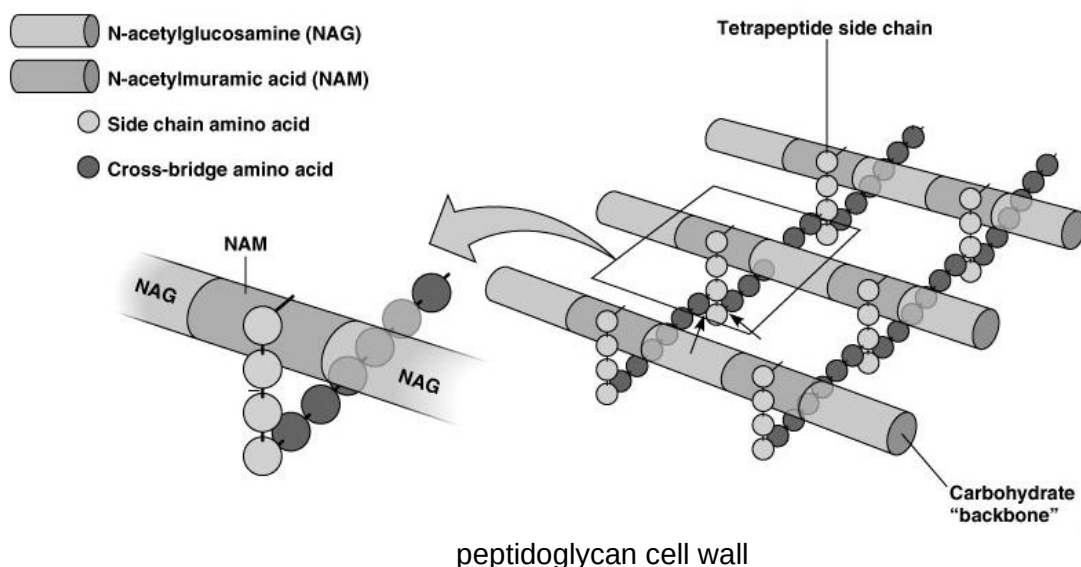
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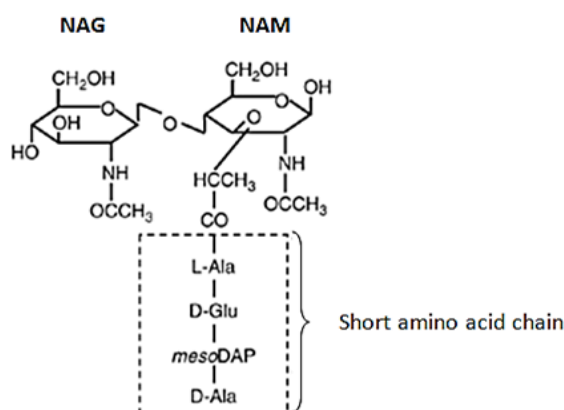
Which student correctly matched the numbered function with the appearance of the cell structure?

	1	2	3
A	V	X	Z
B	Y	W	V
C	Y	X	V
D	V	W	Z

- 4 The peptidoglycan layer in bacterial cell walls is a crystal lattice structure formed from linear chains of two alternating amino sugars, N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM). NAG and NAM differ from glucose only at the C2 and C3 positions.



Each NAM is attached to a short amino acid chain. The figure below shows the molecular structure of NAM and NAG.

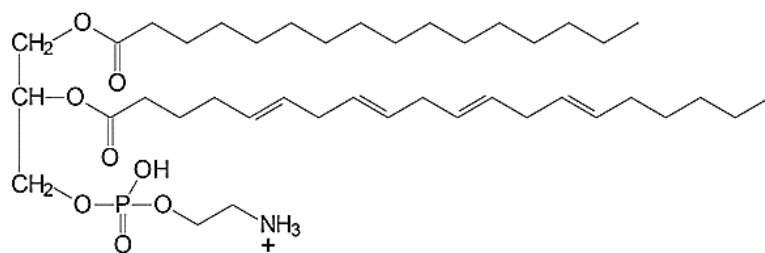


molecular structure of NAG and NAM

Which of the following correctly compares peptidoglycan and cellulose?

	similarity	difference
A	both contain β (1,4) glycosidic bonds	two types of monomers present in cellulose; one type of monomer present in peptidoglycan
B	monomers are rotated 180° to each other	cross-links between cellulose chains are hydrogen bonds; cross-links between carbohydrate chains in peptidoglycan are peptide bonds
C	bundling of chains increases tensile strength	one type of bond found in cellulose fibrils; two types of bonds found in peptidoglycan
D	both comprise chains lying parallel to one another	β -glucose is found in cellulose; α -glucose is found in peptidoglycan

- 5 The structure of phosphatidylcholine, a common membrane phospholipid, is shown.



Which combination correctly describes the synthesis and property of one molecule of phosphatidylcholine?

	number of water molecules eliminated during synthesis	property of phosphatidylcholine
A	3	amphipathic
B	2	amphipathic
C	3	amphoteric
D	2	amphoteric

- 6 Glycine is an amino acid in which the R group is a hydrogen atom.

Which shows the correct sequence of molecules, in order of increasing number of hydrogen atoms?

	increasing number of hydrogen atoms →		
A	α -glucose	glycine	glycerol
B	α -glucose	glycerol	glycine
C	glycerol	α -glucose	glycine
D	glycine	glycerol	α -glucose

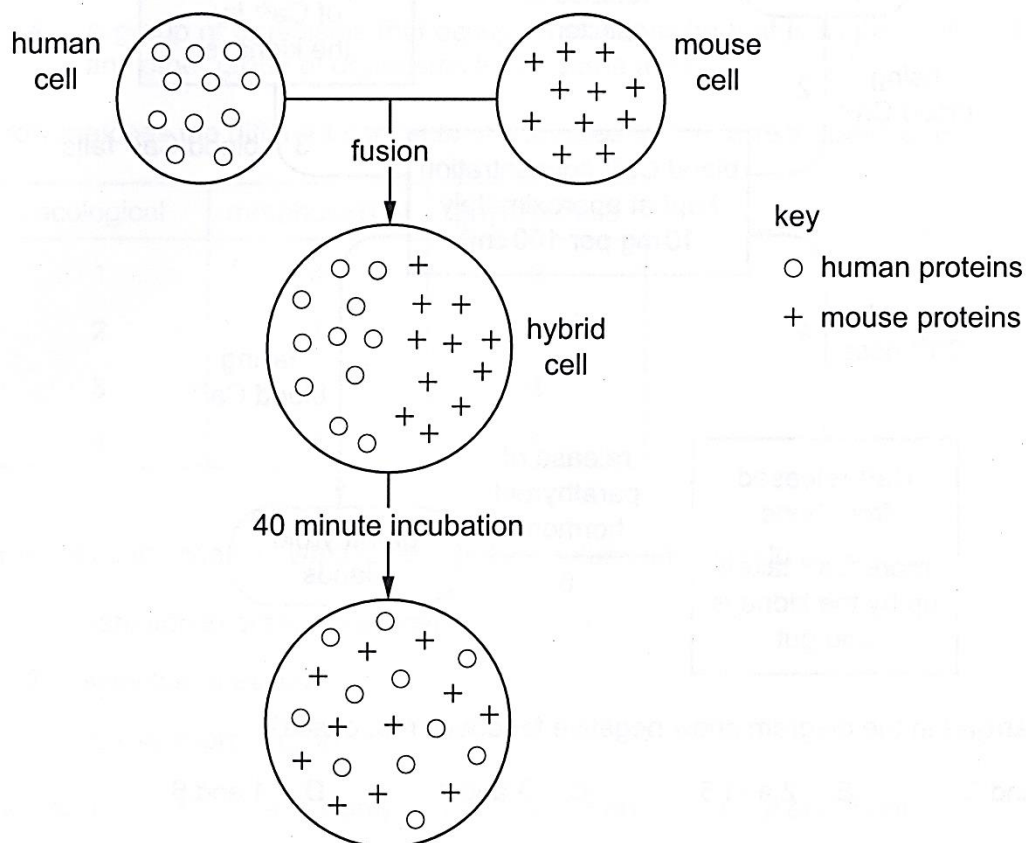
7 Which row about the structure of proteins is correct?

	primary structure	secondary structure	tertiary structure	quaternary structure
A	is the number of amino acids present in a protein	is the right-handed spiral formed by the primary structure	is the result of cross-bonding between specific amino acids in the primary structure	is the sub-unit polypeptides that link together to form a protein
B	is the order of amino acids present in a protein encoded by DNA	is the coiling of a chain of amino acids to form a β -pleated sheet or an α -helix	is the shape formed by the folding of a polypeptide and is held together by hydrogen bonds	contains two types of polypeptide that interact forming the shape of a protein
C	is the result of translation of an mRNA molecule by a ribosome into a chain of amino acids	occurs because of an attraction between hydrogen and oxygen atoms in the peptide bonds	is the result of ionic and hydrogen bonds, disulfide bridges and hydrophobic interactions between amino acids	is formed by four polypeptides and an additional reactive group attached to the protein
D	is the sequence of amino acids in a protein coded by an mRNA molecule	is formed by hydrogen bonding between amino acids forming the primary structure	is formed as a result of interaction of the side chains of amino acids in the primary structure	is formed by the linking together of more than one polypeptide to form a protein

- 8 Human and mouse cells were fused to make hybrid cells. Anti-human and anti-mouse antibodies, carrying different coloured fluorescent dyes, were added. The antibodies bind to the proteins of the cell surface membrane.

The fused cells were incubated for 40 minutes. The locations of the human and mouse membrane proteins were identified at intervals using the fluorescent dyes.

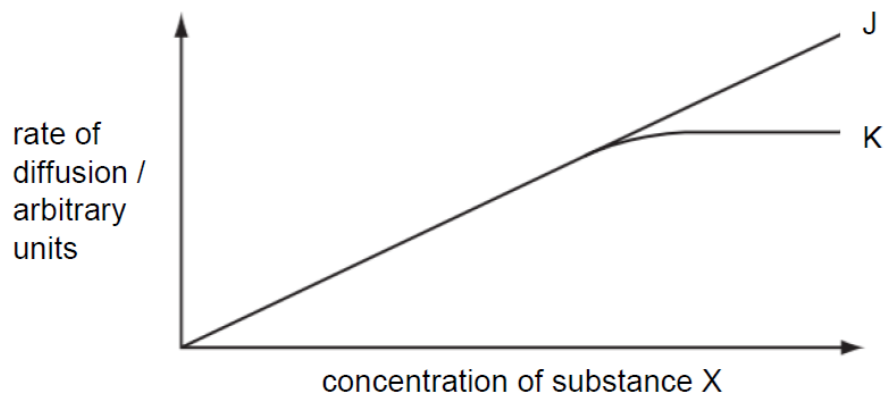
The diagram represents the results of the experiment by showing the positions of the human and mouse proteins on the surface of the cells.



What does this experiment show?

- A The phospholipids of the human and mouse cell surface membranes do not mix.
- B The membrane proteins of human cells can move further than those of mouse cells.
- C Movement of the phospholipids pushes the membrane proteins apart.
- D Some membrane proteins move through the phospholipids to different locations.

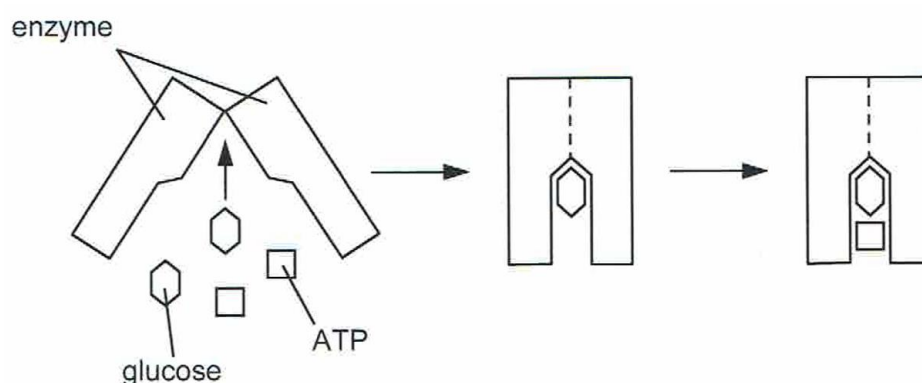
- 9 The graph shows rates of simple diffusion and facilitated diffusion of substance X across the cell surface membrane.



Which row is correct?

	graph K	explanation
A	facilitated diffusion	limited by the number of protein channels in the cell surface membrane
B	simple diffusion	allows the concentrations of substance X inside and outside the cell to be equal in a shorter time
C	facilitated diffusion	allows dynamic equilibrium to be achieved in a shorter time
D	simple diffusion	limited by the number of transient gaps in the cell surface membrane

- 10 The diagram below shows the enzyme hexokinase catalysing the transfer of a phosphate group from ATP to glucose.



Based on the information above and biological knowledge, which of the following correctly describes how hexokinase functions during the reaction?

	increases activation energy of reaction	binds with substrates temporarily	shape of active site changes	enzyme residues undergo reversible changes
A	yes	yes	no	no
B	yes	no	yes	no
C	no	yes	yes	yes
D	no	no	no	yes

- 11 Which statement regarding the enzymatic activity of catalase under the following conditions is correct?

- 1 addition of more hydrogen peroxide
- 2 addition of pH 13 buffer solution
- 3 incubation at 10°C

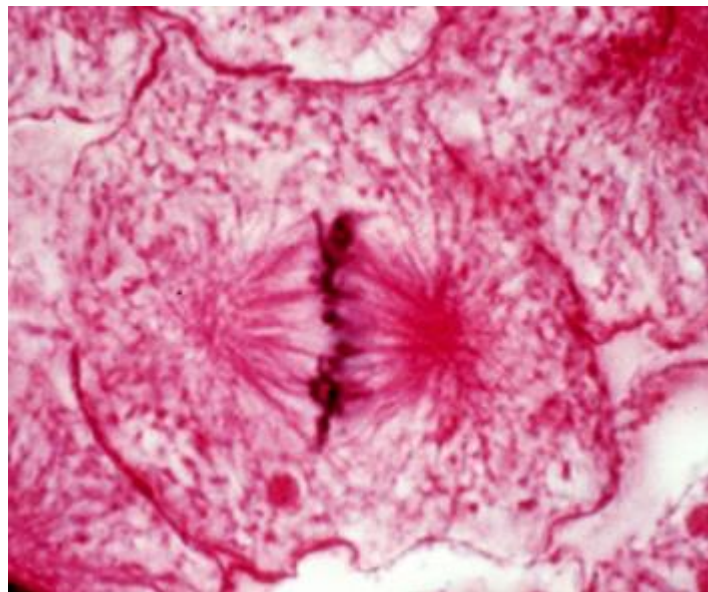
- A** The catalytic and binding residues in all active sites are affected in condition 1, hence the rate of reaction decreases.
- B** The hydrogen and ionic bonds between R groups of residues are broken in conditions 2 and 3, hence the rate of reaction decreases.
- C** The chances of effective collisions to form enzyme-substrate complexes are low in conditions 2 and 3, hence the rate of reaction decreases.
- D** The changes in the 3D conformation of enzymes are irreversible in conditions 1, 2 and 3, hence the rate of reaction decreases.

- 12** The use of induced pluripotent stem cells (iPSCs) overcome the following issues involving the use of other types of stem cells.
- 1 Embryonic stem cells are obtained by irreversibly damaging the blastocyst, which has the potential to develop further.
 - 2 Stem cells are difficult to harvest from an individual as they are present in smaller proportions and are morphologically similar to non-stem cells.
 - 3 The use of foreign embryonic stem cells during treatment may cause harm in the form of tissue rejection to the patient.
 - 4 Although informed consent on the use of embryonic stem cells is given by the blastocyst donor, the potential being has no say.

How many of the issues above are ethical in nature?

- A** 1 **B** 2 **C** 3 **D** 4

- 13** The micrograph below shows a stage of cell division immediately after fertilisation to form a zygote of an Atlantic freshwater fish.



Which of the following sets of terms correctly describes the stage shown above?

- A** DNA amount has doubled, chromosomes are fully condensed
B chromosome number has doubled, cellular poles are established
C microtubules have attached to centromeres, chromatids have separated
D cleavage furrow has formed, chromosomes are found at a single location

- 14** Diploid ($2n$) organisms that reproduce sexually produce haploid (n) gametes.

Some plants, such as wheat, can produce diploid or haploid gametes. These gametes can fertilise other diploid or haploid gametes.

Which statements are correct for plants like these?

- 1 Diploid gametes may be produced by a fault in the reduction division (meiosis).
- 2 The offspring will always show an increased chromosome number.
- 3 The offspring could be either $2n$, $3n$ or $4n$.
- 4 The chromosome number could, in theory, increase with each generation.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1, 3 and 4 **D** 2, 3 and 4

- 15** What are the products when a DNA molecule replicates?

- A** two molecules of DNA each made of a sequence of paired bases
- B** two molecules of DNA each made of a sequence of paired nucleotides
- C** two strands of DNA each made of a sequence of paired bases
- D** two strands of DNA each made of a sequence of paired nucleotides

- 16** Which of the descriptions of RNA is incorrect?

- A** Ribosomal RNAs catalyse the formation of phosphodiester bonds.
- B** Transfer RNAs bind with messenger RNA during translation.
- C** Small nuclear RNAs bind with proteins to form spliceosomes.
- D** Messenger and transfer RNAs can contain secondary structures.

17 Two students were discussing the involvement of DNA and RNA in transcription and translation.

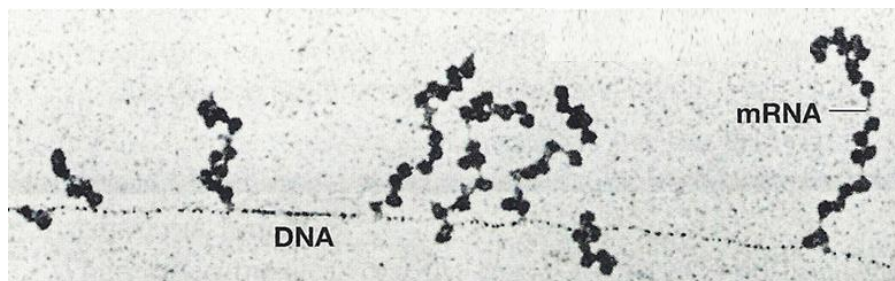
- Student 1 always stated correct facts.
- Student 2 gave further information, which was sometimes correct.

	correct facts given by student 1	further information given by student 2
1	A length of mRNA is 747 nucleotides long, including stop and start codons.	This mRNA can produce a polypeptide that is 249 amino acids long.
2	Adjacent mRNA codons of AAU and CUG bind to complementary tRNA anticodons.	There is a total of 14 hydrogen bonds formed between these two codons and their anticodons.
3	RNA polymerase catalyses the formation of the mRNA from the template strand of DNA.	During translation, an RNA adenine nucleotide will pair with a DNA thymine nucleotide.
4	A DNA adenine nucleotide is structurally different to an RNA adenine nucleotide.	The difference is in the hexose sugars; DNA is deoxyribose and RNA is ribose.

Which further information, given by student 2, is correct?

- A** 1 and 4 **B** 2 and 3 **C** 2 only **D** 4 only

18 Protein synthesis in a prokaryote is shown in the micrograph below. Several mRNAs are evident.

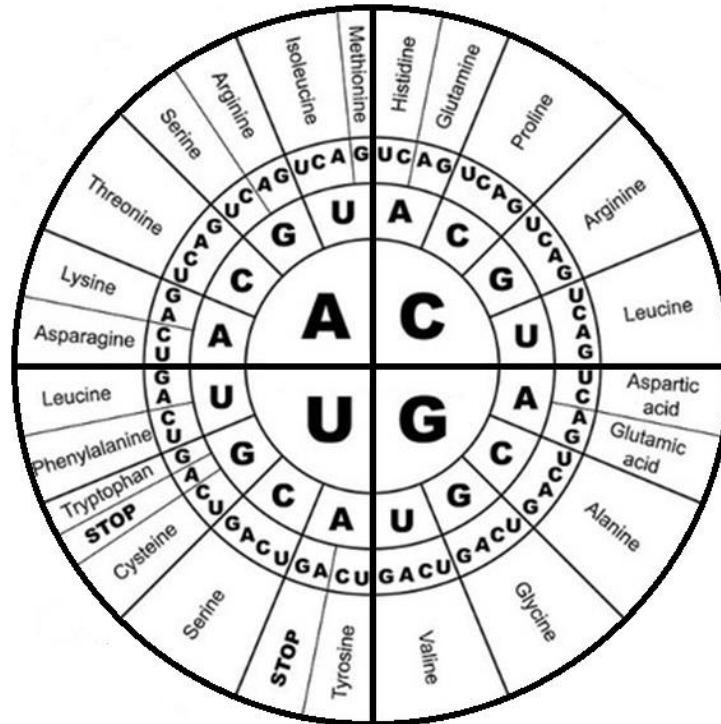


How many of the following can be determined in the micrograph above?

- 1 the locations of ribosomes
- 2 the locations of RNA polymerases
- 3 the locations of the 5' ends of the mRNAs
- 4 the locations of polypeptides undergoing synthesis

- A** 1 **B** 2 **C** 3 **D** 4

19 The mRNA codons for amino acids are shown below.



A mutagen causes the adenine in DNA to pair with cytosine during transcription.

Which tripeptide will be synthesised when the DNA fragment ACCGTCAAT is used in protein synthesis in the presence of this mutagen?

- A proline-valine-proline
- B arginine-glutamine-proline
- C tryptophan-glutamine-leucine
- D threonine-valine-asparagine

20 Aneuploidy is a form of chromosomal aberration.

Which of the following correctly describes aneuploidy?

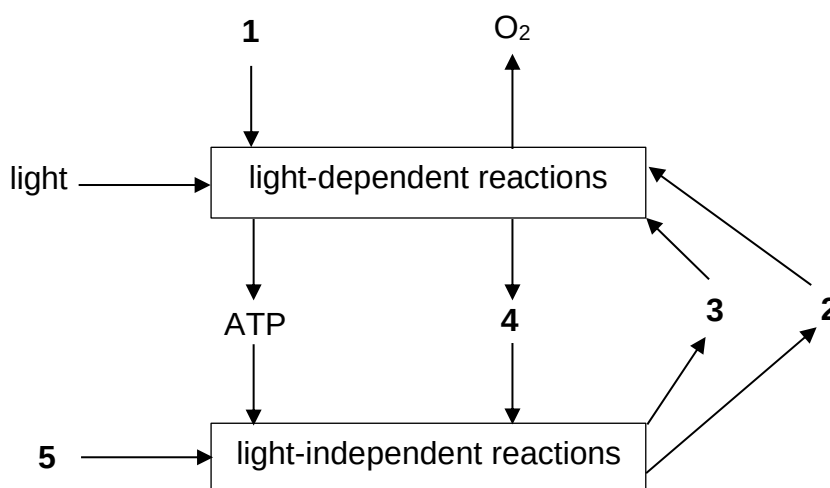
- A It can arise due to gene mutations that affect kinetochore proteins.
- B It can result due to the complete failure of cytokinesis after nuclear division.
- C A zygote that contains two X chromosomes, instead of one X chromosome and one Y chromosome, is affected by aneuploidy.
- D In individuals affected by aneuploidy, translocation of chromosome segments is commonly observed.

- 21** Melanoma is a form of skin cancer that begins in cells called melanocytes that control pigment production in the skin.

Which of the following correctly compares a cancerous and a normal melanocyte?

	similarity	difference
A	both cells arise from pre-existing cells	the cell cycle of cancerous melanocytes is arrested at either the S phase or M phase
B	some genes are switched off in both cells	cancerous melanocytes continue to divide even when they are tightly packed
C	both cells contain well-defined nuclei	cancerous melanocytes are always in the M phase of the cell cycle
D	both cells are specialised	more gene mutations are evident in cancerous melanocytes

- 22** The diagram summarises the process of photosynthesis.



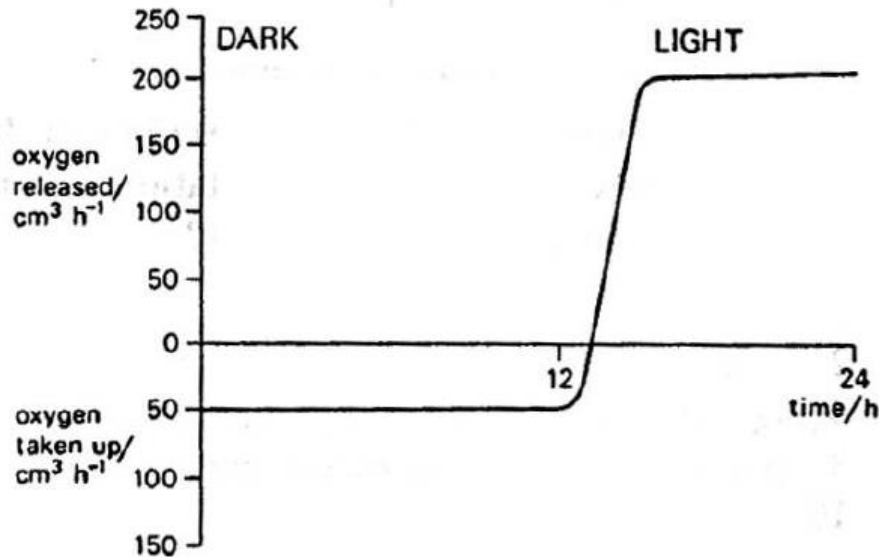
Which row identifies the reactants **1**, **2**, **3**, **4** and **5**?

	1	2	3	4	5
A	carbon dioxide	ADP and phosphate	reduced NAD	NAD	water
B	carbon dioxide	reduced NADP	ADP and phosphate	NADP	water
C	water	NAD	reduced NAD	ADP and phosphate	carbon dioxide
D	water	NADP	ADP and phosphate	reduced NADP	carbon dioxide

- 23** Data was obtained relating to the rates of oxygen release and uptake in plants.

The plants were placed in the dark for 12 hours followed by 12 hours in the light. Temperature was kept constant throughout the experiment.

The results are shown in the graph below.



Which of the following is the most accurate estimate of the total volume of oxygen used by the plant for respiration during the 24 hours of the experiment?

- A** 600 cm^3 **B** 1200 cm^3 **C** 1800 cm^3 **D** 3000 cm^3

- 24** The following compounds occur in cellular respiration.

- 1 pyruvate
- 2 glucose
- 3 acetyl coA
- 4 oxaloacetate
- 5 carbon dioxide

Which shows the correct order in which these compounds first occur during cellular respiration?

- A** 2, 1, 3, 4, 5
B 2, 1, 5, 3, 4
C 2, 3, 1, 5, 4
D 2, 3, 1, 4, 5

- 25** Your teacher has never had much energy. He goes to a doctor for help and is sent to the hospital.

After some laboratory tests, it was discovered that his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal.

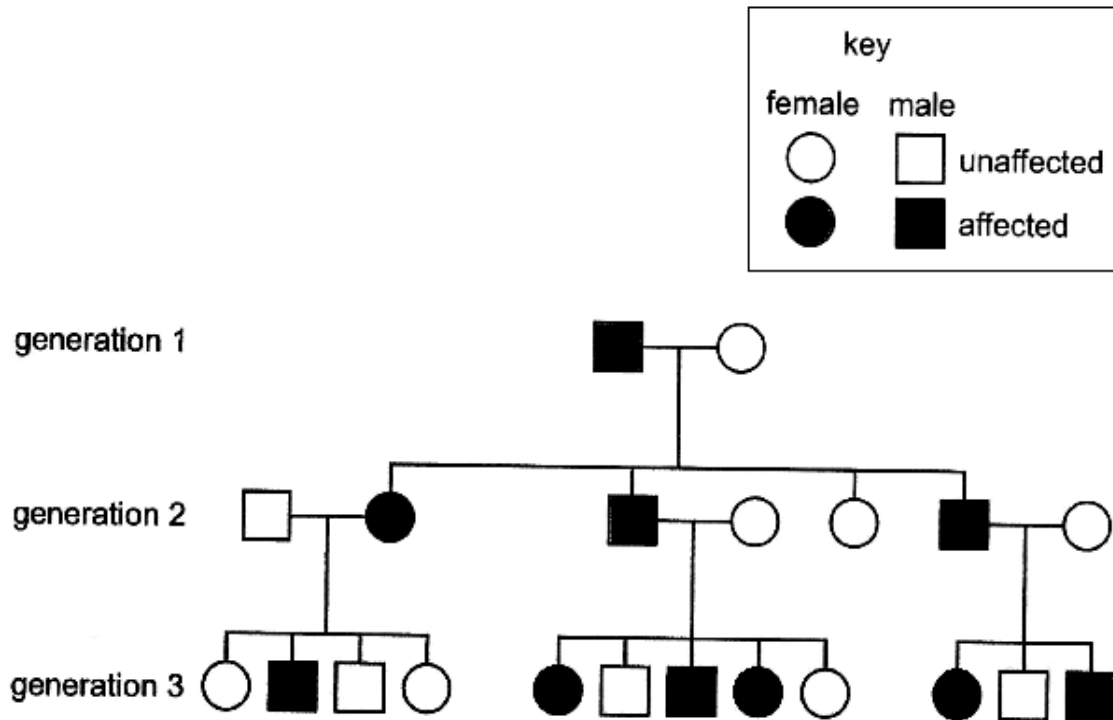
Which of the following is the best explanation for his condition?

- A** His diet is excessively high in lipids and proteins, and low in carbohydrates.
 - B** His cells contain an inhibitor that prevents his mitochondria from using oxygen.
 - C** The transport proteins on the envelopes of his mitochondria are defective.
 - D** Reduced hydrogen acceptor molecules from glycolysis cannot enter his mitochondria.
- 26** In humans, the gene for brown eyes is dominant to the gene for blue eyes and identical twins occur once in approximately 300 births.

What is the probability of a couple, heterozygous for brown eyes, having identical twins with brown eyes?

- A** 1 in 1200 **B** 1 in 800 **C** 1 in 400 **D** 1 in 300

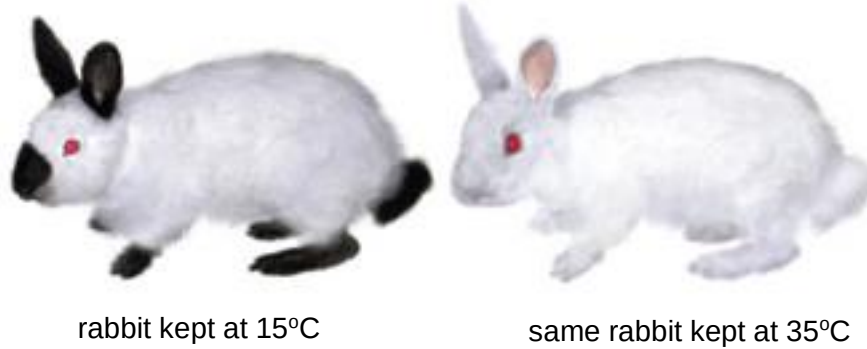
- 27 The pedigree shows the inheritance of a dominant genetic condition in a family for three generations.



Which evidence indicates that this genetic condition is autosomal?

- A Affected females always have affected sons.
- B Affected males do not pass it on to their sons.
- C Affected parents always have affected offspring.
- D Males and females are equally affected.

- 28** The figure below shows the outcome of keeping a Himalayan rabbit at 15°C for a period of six months, followed by keeping the same rabbit at 35°C for another period of six months.



From the information above and biological knowledge, which of the following best explains the observed outcome?

- A** Fur colour in Himalayan rabbits is only affected by temperature.
- B** The genes present and the environment have effects on the fur colour of Himalayan rabbits.
- C** The production of white fur in Himalayan rabbits is controlled by an enzyme that is temperature-sensitive.
- D** An increase in temperature causes gene mutations which prevent black fur from forming in Himalayan rabbits.

Use the following information for Questions 29 and 30.

The Norway rat (*Rattus norvegicus*), a widespread pest, was controlled in the past by the anticoagulant warfarin. This chemical substance inhibits the clotting of blood.

Resistance to the widely used poison warfarin is now extremely common in rats. Warfarin inhibits an enzyme in the liver that is necessary for the recycling of vitamin K. Vitamin K is involved in the production of substances required for blood clotting.

Normal rats fed on warfarin suffer a fatal haemorrhage. Resistant rats apparently do not use vitamin K in the same way and maintain normal blood clotting times, even when they have eaten large amounts of warfarin.

Resistance to warfarin in rats is determined by a single autosomal dominant allele, **R**.

The table below indicates the response of individuals to warfarin, vitamin K dependence and relative reproductive fitness. The highest fitness level is 1.00.

characteristic	genotype of individuals		
	rr	Rr	RR
response to warfarin	susceptible	resistant	resistant
vitamin K dependence	no	intermediate	high
relative fitness	0.68	1.00	0.37

29 Which of the following statements about the allele for resistance to warfarin is correct?

- A It resulted from a mutation caused by warfarin.
- B The phenotype resulting from it is favoured by natural selection.
- C It confers a selective advantage in all environments.
- D Its frequency in the population increased because it is the dominant allele.

30 In a population where warfarin is no longer applied, the most likely outcome is that the

- A frequency of the dominant allele will be reduced.
- B proportion of recessive individuals will gradually decline.
- C dependence of the population on vitamin K will increase.
- D heterozygotes will have a reproductive advantage over the homozygotes.

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